BUILT — Making it POP: real-time WebGL 3D + dynamic schematics (v0.40.0)

On the current RTX 4050, fully offline (no Three.js / no CDN). Grounded: same geometry & pages, just rendered & navigated dynamically.

3D VIEWER — engine/ui/gl3d.js (dependency-free WebGL)

- **Real-time WebGL render**
  A tiny self-written GL renderer (no library) draws the FLIS-scaled shapes lit in 3D — replaces the flat SVG. Offline + RPS-safe (SVG fallback if WebGL is missing).
- **Glossy multi-light shading**
  Key + fill + rim/fresnel lights, sharp specular, soft tonemap — a metallic 'product render' look that pops.
- **Antialiasing + smooth surfaces**
  MSAA on; round families (cyl/hex/disc) use averaged normals so cylinders look round, boxes stay crisp.
- **Idle turntable**
  Auto-spins gently; grab to orbit (pauses), release and it resumes after a beat — dynamic by default.
- **Orbit · zoom · reset**
  Drag to orbit, scroll to zoom, double-click / ⟲ to reset. Wired into the 3D Library; the cart viewer adopts it next.

Grounded: representative shapes from real dimensions — better rendering, nothing invented.

SCHEMATICS VIEWER — dynamic navigation

- **Buttery pan + cursor zoom**
  Drag to pan; scroll zooms toward the point under the cursor (transform-origin follows the pointer). Smooth, no reload — pure transform.
- **Blueprint mode**
  One tap flips a page to white-lines-on-blue (invert + hue) — classic schematic look, instant 'pop'.
- **Clean (legibility)**
  Server-side grayscale + contrast + de-speckle on the real page; toggles live.
- **Fade page transitions**
  Pages cross-fade in (pre-loaded) instead of blanking — feels fluid.
- **Reset · arrow-key paging**
  ⟲ resets zoom+pan; ←/→ page through; double-click resets. All on the real rendered pages.

Grounded: it's the real PDF page — rendered, themed, and navigated, never altered.

WHY IT STAYS HONEST + OFFLINE

All of this is presentation only (R1/R6): the dataset, search, and 104th sheet are untouched. The 3D is the same parametric geometry from real FLIS dimensions, just lit properly; the schematic is the same real page, just navigated and themed. No Three.js and no CDN — the WebGL renderer is ~120 lines we own, so the app stays fully offline and the SVG fallback keeps Win7/Vista (RPS) working.

Next envelope steps (per proposal 55): parametric template library (thread helix/gear teeth), schematic vectorisation + per-wire hover — both on the 4050. Photogrammetry + net-extraction await the desktop/RTX 5070 (reminder set).

BUILT diagram. Dark (R3). v0.40.0 · 2026-06-02 · new: engine/ui/gl3d.js + /gl3d.js route.